Assignment 3:

- This assignment exposes students to enhanced entity relationship modeling techniques using both Chen and Information Engineering notation as well as database implementation from a model and relational algebra.

- Due date is published on Nexus for this Assignment.

Software / Diagramming
What ERD diagramming tools can you use for this assignment?

- You can hand-draw your models, but they must be neat/tidy and perfect notation.
- For Chen Notation you have two choices:
  - Free ACS Diagramming Tool (use Chrome browser for best results): https://courses.acs.uwinnipeg.ca/erd/DrawERDSept2023.html
  - Free online tool ERDPlus: https://erdplus.com/
- For Information Engineering Notation you must use Lucidchart.

  The special subtype symbol ( ) needs to be added to your toolset. See the tool primer issued at the beginning of this course for details.

Checklist: What to hand in?

Part A:

- ☐ A PDF of your completed Chen EERD model

Part B:

- ☐ A PDF of your completed Information Engineering Model

Part C:

- ☐ A single executable database script containing your queries. A script template is provided.

Part D:

- ☐ A submission template has been provided for you with this assignment for Part D. Any drawings or images for this section must be placed in the template. **IMAGES UPLOADED INDEPENDENTLY WILL NOT BE ACCEPTED.**

IMPORTANT

- Significant deductions may apply for scripts that fail to execute without errors. Ensure you test your work.
- Scripts without your name and student number may also receive a deduction.

Part A: Enhanced ERD Modeling [40 marks]

Develop a data model using Chen EERD that meets the requirements for a simplified car rental application. You may either hand-draw the model or use the free diagramming tool from ACS at the following link (https://courses.acs.uwinnipeg.ca/erd/DrawERDSept2023.html). A clear, easy-to-read PDF version of your diagram must be submitted.

Your design must:

- include at least one suitable use of each of the following:
  - a weak relation and weak entity.
  - Exclusive (aka disjointed) sub-type.
  - inclusive (aka overlapping) sub-type.
- indicate cardinality on each relationship.
- model all attributes
- identify any unique attributes within an entity.
- Must be reasonably neat and easy to read/follow.

Data Requirements for the iRent My Vehicle Application

Overview: This application will allow owners of vehicles to rent their vehicle out to earn revenue when their car is not needed or used frequently. By creating a pool of vehicles available for rent, the application will have the potential to provide more affordable rental alternatives compared to commercial companies. Requirements have been greatly simplified to reduce the complexity for this portion of the assignment.

Requirements Checklist

User

- ☐ Users of the system can register a single account which allows them to act as a renter (the person who offers a vehicle for rent) or a lessee (the person who rents the vehicle). Each user has a username, first, last, and email address. Records for renters include an insurance policy number and insurance company name while lessees must provide a driver's license number and license expiry date.

Vehicle

- ☐ Each vehicle is described by the following attributes: A vehicle number (unique, assigned by the system), a plate number, the make (e.g., Honda), and model (e.g., Civic).
- ☐ The system supports different types of vehicles including motorcycles, regular passenger cars, trucks, and recreational vehicles (e.g., motor homes). Unique attributes are kept for motorcycles, trucks, and recreational vehicles as follows:
  - For motorcycles we record the recommended minimum height of the rider.
  - For trucks, we record the towing capacity (how much weight can be pulled).
  - For RVs we record how many people can sleep in it.

Rental Agreement

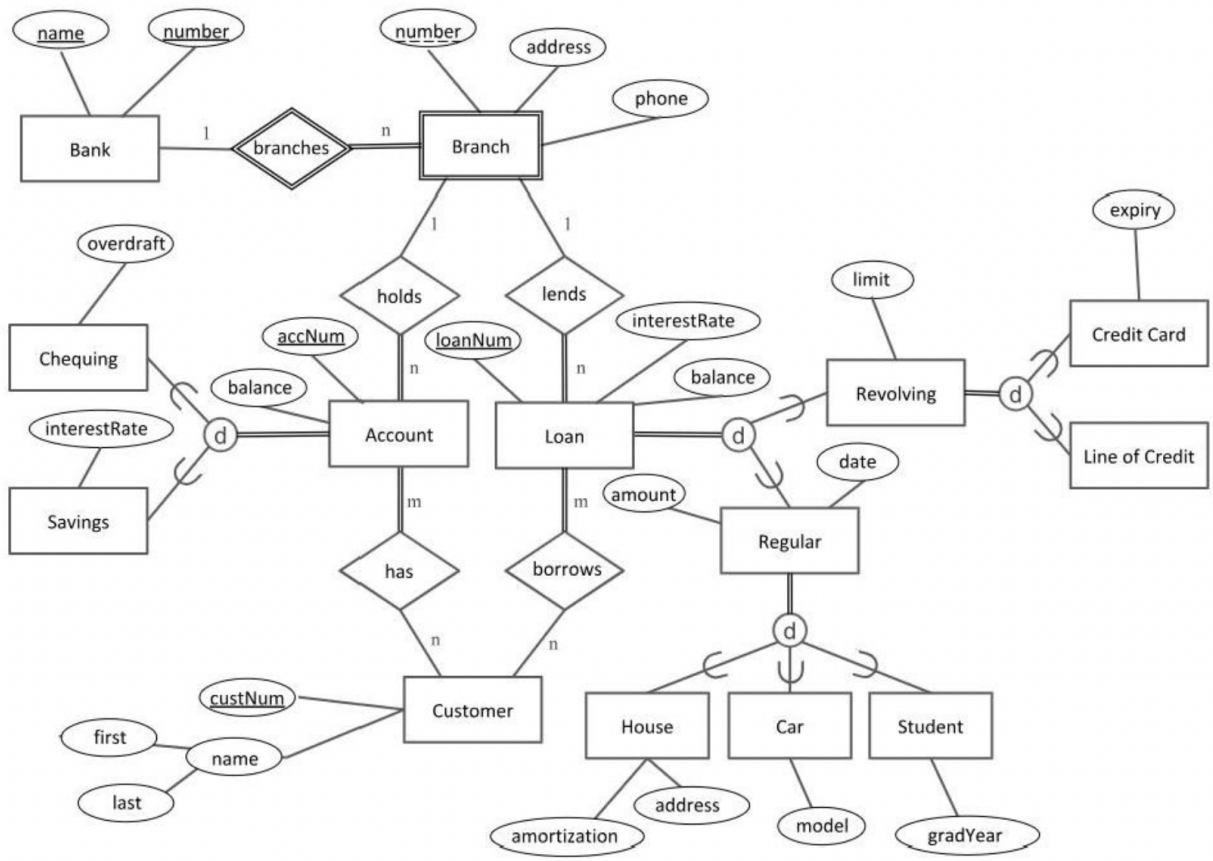
- ☐ A rental agreement is identified by a unique agreement number and includes the start date and end date of the rental and the fee charged.
- ☐ Only one vehicle can be rented for each agreement. A vehicle may be rented several times over the course of time.
- ☐ Each rental agreement must identify one renter and one lessee. Over time, renters and lessees can participate in many agreements.

Payments/Refunds

- ☐ Our application recognizes different types of payments. A reservation deposit (to reserve a vehicle), a damage deposit, a refund, and a normal rental payment. Each payment or refund is identified by a unique transaction ID. We also record the date of the transaction and the method of payment used.
- ☐ Each transaction must relate to a single rental agreement. A rental agreement may incur many transactions.

Part B: Translating Modeling Notation from Chen to Information Engineering (15 marks)

Consider the following Chen EERD for a simplified banking system.

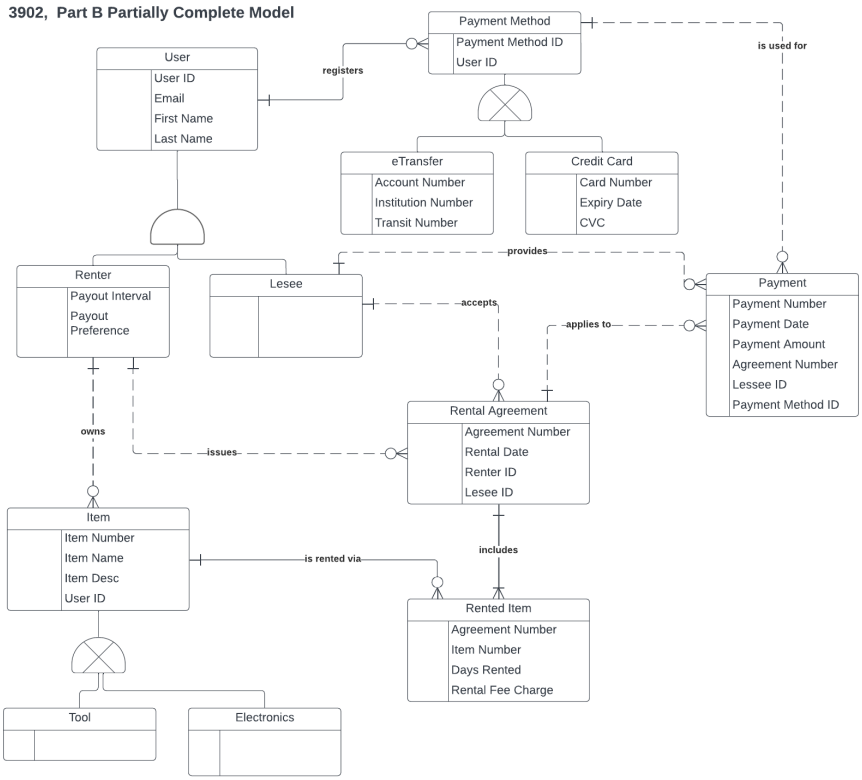


Create an equivalent logical data model in Information Engineering notation. Your completed model must meet the following criteria:

- Must express the same attributes, entities, and structures accounted for in the Chen notation.
- Should not have extra or unnecessary entities or attributes. You may need to add attributes to eliminate composite (non-atomic attributes) such as "address".
- Must adhere to the Information Engineering notation standards described for this course in the appendix (and also available here at this Lucid Chart Link)
- Must distinguish identifying and non-identifying relations.
- Must be normalized to BCNF.
- Must identify all PK and FKs (no AKs are required).
- You do not need to identify data types or null/not-null designations for this model.

Part C: Database Implementation from a Logical Data Model (15 marks)

An online copy of [this diagram is also available on Lucidchart](#).



Prepare a Postgres DDL Script that implements a physical database from the model above.
The logical model is partially complete (no PK or FK declarations are present).

Implementation Requirements

- You must decide what the PKs and FKs should be.
- Primary and foreign key declarations must agree with the cardinality shown in the diagram above.
- The sample data (next page) will assist with data type selection and PK/FK designation.
- You must decide how best to represent the sub-type structures in the physical implementation.

Your DDL must be a single executable script that includes:

- All create table statements.
- Reasonable Postgres Data Types (based on the sample data provided) with not null constraints where applicable.
- Declaration of all PKs and FK constraints (no RI actions or AKs are needed for this question).
- A script template is provided with the assignment.
- Insert statements to load the sample data shown below.

Part C: Sample Data

- Underlined items indicate unique attributes or combinations of attributes.
- Notice that the data for the sub-types is shown separately.
  Depending upon your final table implementation, you may need to adapt the structure of the data to suit your design.

User

User ID	Email	First Name	Last name
cjoyal	Claude.Joyal@gmail.com	Claude	Joyal
srobins	selina.robins@hotmail.com	Selina	Robins

Renter

User ID	Payout Interval	Payout Preference
cjoyal	Monthly	Cheque
srobins	Weekly	Direct Deposit

Payment Method

Payment Method Number	User ID
87	cjoyal
88	cjoyal
65	srobins

eTransfer

Payment Method ID	Account Number	Institution Number	Transit Number
87	10005007	097	50198
65	10889746	003	84752

Credit Card

User ID	Card Number	Expiry Month	CCV
88	45000080000581163	2024-09	887

Payment

Payment Number	Payment Date	Payment Amount	Agreement Number	Lessee ID	Payment Method ID
8899	2024-02-14	80.50	7465	srobins	87

Item

Item Number	Item Name	Payout Preference	User ID
1348	PS5	Playstation5	cjoyal
1349	Cordless Drill	9V Dewalt	cjoyal

Rental Agreement

Agreement Number	Rental Date	Renter ID	Lessee ID
7465	2024-02-13	cjoyal	Srobins

Rented Item

Agreement Number	Item Number	Days Rented	Rental Fee Charge
7465	1348	1	40.50
7465	1349	1	40.00

Part D: Relational Algebra (15 marks)

D.1 **[5 marks]** Write a single SQL statement that is equivalent to the logic expressed by the following relational algebra.

EXP_STAFF(id,,Last) ← $\pi_{(emp_id,\ lname)}\ \sigma_{years_experience>=5}$ (EMPLOYEE)

SAL_CHG(id,PayRaise,CurrentSal) ← $\pi_{(emp_id,\ ann_sal * 0.05,\ ann_sal)}\ \sigma_{salary_is_active= true}$ (SALARY)

RESULT (ID, Last, NewSalary, OldSalary, Raise) ← $\pi_{id,\ Last,\ PayRaise + CurrentSal,\ CurrentSal,\ PayRaise}$(EXP_STAFF ⋈ SAL_CHG)

D. 2 **[5 marks]** Write the relational algebra expression(s) for the following SQL statement.

```
Select first_name, last_name, t.country_visited
        FROM person p
                JOIN trip t ON (p.person_id = t.person_id)
WHERE t.trip_start_date >= '2024-01-01' and t.trip_purpose = 'Pleasure';
```

D. 3 **[5 marks]** Prepare an optimized query tree for the following SQL Statement given the following assumptions:

- The order table referenced in the query is very large with over 1M rows of data.
- The country table related to the customer is very small (we only have customers in 4 countries).
- The customer table is moderately large (we have ~10,000 customers).

```
Select first_name, last_name, c.country, sum(revenue)
        FROM customer c
                LEFT OUTER JOIN country n ON (c.country_id = c.country _id)
                JOIN order o ON (c.customer_id = o.customer_id)
WHERE c.customer_status = 'ACTIVE' and
                n.country_name = 'Canada' and
                o.order_date between '2024-01-01' and '2024-03-31' ;
```

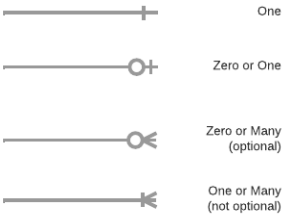
Information Engineering Standards (Also Available Online at this Lucid Chart Link)

Logical Data Types

Logical Data Types		
	Numeric, no decimal places	Integer
	Numeric with high precision decimal places	Float
	Numeric where x=total # digits, y=number of decimal places. (e.g., 1500.25 would be Numeric(6,2))	Number(x,y)
	A text value where n specifies the maximum number of characters.	Text(n)
	A true/false or yes/no value	Boolean
	An unlimited text field suitable for long descriptions.	Memo
	A binary object such as a image, file, or video.	Binary
	A date value with time	Datetime
	A date value without time	Date
	A time value without date	Time

Relationship Type

Relationship Cardinality



Mandatory Attributes

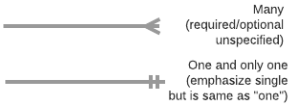
Student		
PK	Student Number	Integer
	First Name	Text (50)
	Last Name	Text (50) : Not Null
	SIN	Integer : Not Null

Specifying that an attribute is mandatory is commonly done by adding a colon with "Not Null".

PK and Not Null
It is permissible to omit the "not null" indication for attributes that are included in the primary key since any member of a primary key is deemed mandatory by default.

Relationship Cardinality (Not to be Used)

(Supported by Lucidchart and other tools but these are either ambiguous and/or redundant)



Sub-Type Structure

Keys

Simple (Single Attribute Keys)		
PK	Indicates a primary key	
AK	Alternate Key	
FK	Foreign Key	

Compound Keys (Example)

Student		
PK	School Number	
PK	Student Number	
	First Name	
AK1.1	Last Name	
AK1.2	Postal Code	
AK2	SIN	

The notation for compound keys is shown here with alternate keys as an example.

The Primary Key is Made up of two Attributes
Since we can only have one primary key per entity, no numbering is needed.

Alternate Key Numbering
Numbering can be used to distinguish two different alternate keys in the same entity.

AK1 is an laternate key that includes two attributes
AK1.1 Last Name (the first member of AK1)
AK1.2 SIN (the second member of AK1)

AK2 is a second, independent alternate key that has only the one attribute.

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Attributes that Participate in Multiple Keys (example)

Phone		
PK, FK	CustomerNumber	
PK	PhoneType	
	PhoneNumber	
	Extension	

In this example, the primary key is a combination of customer number and phone type. The customer number is also a foreign key. We use a comma-seperated notation to indicate the duality of customer number.

Customer		
PK	Customer Number	
AK1.1, AK2.1	Last Name	
AK1.2	Postal Code	
AK2.2	Date of Birth	

In this example, last name is part of two alternate keys. AK1 includes last name and postal code while AK2 includes last name and date of birth. The two keys are independent but share some common attributes (last name) in this case.